DAY10-LINUX-FILES

THEORY (answer all 20 questions in your own words):

1. What is a Linux directory?

A directory is a folder used to organize and store files and other directories.

2. What does mkdir do: Creates new directory

3. What does touch do: Create Empty file

4. What does cat do: Opens a file

5. What does cp do: Copys a file

6. What does mv do: Moves or renames a file.

7. What does rm do: Removes a file.

8. What does ls -l display: file permission, owner, size etc..

9. Explain r, w and x permissions: r for read, w for write and x for execute.

10. Explain user, group and others.

- User: The owner of the file.
- Group: Users belonging to the file's assigned group.
- Others: All other users on the system.

11. What does chmod do: Change mode

12. What does chmod 755 represent: change the full access to owners and read/access to everyone.

13. What does chmod 644 represent: only owner has the permission to write or modify others can only read.

14. What is file ownership: Identifies the user and group that own a file or directory.

15. What is the purpose of /etc?

The /etc directory in Linux is the central repository for all system-wide configuration files. It controls how the system, hardware, and installed applications behave.

16. What is the purpose of /home?

The /home directory in Linux acts as the central repository for user personal data and files.

17. What is commonly stored under /var: Log files, caches, and other frequently changing data.

18. Why should rm be used carefully: Because it removes file and directories which may be hard to restore or recover.

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19. Why are Linux permissions important?

They control who can read, write, or execute files and directories.

20. How can incorrect permissions create a security risk?

Tasks:

SETUP: cd ~ · mkdir day10_lab · cd day10_lab · pwd (verify you are inside the lab directory)

```
┌──(kirthikaa㉿kali)-[~]
└─$ mkdir day10_lab

┌──(kirthikaa㉿kali)-[~]
└─$ cd day10_lab

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ pwd
/home/kirthikaa/day10_lab
```

TASK 1 — Create files: touch notes.txt · touch report.txt · ls → you should see both files.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ touch notes.txt

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ touch report.txt

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls
notes.txt  report.txt
```

TASK 2 — Create directory: mkdir evidence · ls.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ mkdir evidence

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls
evidence  notes.txt  report.txt
```

TASK 3 — Add content: echo "Jayvik Labs Day 10 Linux Lab" > notes.txt → cat notes.txt. The > writes command output into a file (lab files only).

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ echo "Jayvik Labs Day 10 Linux Lab" > notes.txt

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ cat notes.txt
Jayvik Labs Day 10 Linux Lab
```

TASK 4 — Copy: cp notes.txt backup.txt · ls → backup.txt exists.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ cp notes.txt backup.txt

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls
backup.txt  evidence  notes.txt  report.txt
```

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TASK 5 — Move: mv report.txt evidence/ · ls evidence → report.txt moved in.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ mv report.txt evidence/

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls evidence
report.txt
```

TASK 6 — Rename: mv backup.txt notes-backup.txt · ls.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ mv backup.txt notes-backup.txt

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls
evidence  notes-backup.txt  notes.txt
```

TASK 7 — View permissions: ls -l notes.txt → identify Owner / Group / Others permissions.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls -l notes.txt
-rw-rw-r-- 1 kirthikaa kirthikaa 29 Oct 14 16:21 notes.txt
```

TASK 8 — Change permissions: touch demo.sh · ls -l demo.sh · chmod u+x demo.sh · ls -l demo.sh → what changed?

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ touch demo.sh

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls -l demo.sh
-rw-rw-r-- 1 kirthikaa kirthikaa 0 Oct 14 16:26 demo.sh

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ chmod u+x demo.sh

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls -l demo.sh
-rwxrw-r-- 1 kirthikaa kirthikaa 0 Oct 14 16:26 demo.sh
```

- The owner gained execute permission

TASK 9 — Numeric permissions: chmod 644 notes.txt · ls -l notes.txt → explain Owner r/w, Group r, Others r.

```
┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ chmod 644 notes.txt

┌──(kirthikaa㉿kali)-[~/day10_lab]
└─$ ls -l notes.txt
-rw-r--r-- 1 kirthikaa kirthikaa 29 Oct 14 16:21 notes.txt
```

- Owner (6): Read (r) + Write (w)
- Group (4): Read (r) only
- Others (4): Read (r) only

Owner has access to read, write and modify file. But others can only read the file

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TASK 10 — Safe delete: touch delete-me.txt · ls · rm delete-me.txt · ls. LAB FILE ONLY — always verify the name before rm.

